

ISEP — Digital Engineering School

Compensation project 2025/2026 · 5 ECTS · due 31 August 2026

Gym Buddies

An individual web application that connects athletes so they can train together, motivate each other, and find a gym buddy.

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This report **summarizes** the wiki in gym-buddy-documentation (**1.2.0, 2026-09-14**). It is not a second specification. Functional and technical rules live in that repository.

1. Introduction and academic framing

The assignment covers Software Engineering, Web Technologies, and Algorithms and Advanced Programming. Work is individual. Defense is 20 minutes plus 30 minutes of questions. The instructor account `maurras.togbe@isep.fr` must be a collaborator on every private repository.

Repositories

Repository	Role	URL
gym-buddy-documentation	Wiki, tickets, Gym Buddy Project	Open repository
gym-buddy-openapi	OpenAPI 3.1 \$ref tree (HTTP source of truth)	Open repository
gym-buddy-service	Java 25 LTS / Spring Boot API	Open repository
gym-buddy-ui	Angular 22 member app + /admin	Open repository

2. Problem and users

Gym Buddies helps athletes find a training partner by combining a friends graph, public/private profiles, capacity-limited sessions, and explainable “why this person” suggestions. Members use the social application; staff get a separate back-office bundle. Native mobile, payments, and wearables are out of scope.

3. Functional overview

Area	FS prefix	Required behavior
Accounts / JWT	FS-ACCT	Register, login, refresh, logout, password, close
Profiles	FS-PROF	Public / private, stub for strangers
Friends	FS-FRND	Request, accept, block
Feed / posts / comments	FS-FEED / POST / CMT	Friends feed, likes, depth 4
Events	FS-EVT	Instant / WEEKLY, apply, transactional capacity
Search / suggestions	FS-SRCH / SUGG	People+events; FoF generate-and-score
Messaging / media	FS-MSG / MED	Text, image, audio; SeaweedFS signed URLs
Admin / fixtures	FS-ADM	Roles, hide, audit; Datafaker seed 20260813

Highlights for the defense: the friends news feed (FS-FEED), friends-only events with a last-seat CONFLICT (FS-EVT-07), and suggestion cards that print a primary reason (FS-SUGG-03).

4. Architecture and data model

Gym Buddies is a **modular monolith** behind one HTTP API and a WebSocket gateway. Two Angular clients (member and /admin) share that API. Bounded contexts (auth, social, events, search, chat, media, admin) live in one deployable. Microservices were rejected: operational cost dwarfs the academic benefit for one student.

Runtime: PostgreSQL 18 (system of record), Redis (refresh credential state and messaging pub/sub), SeaweedFS (private images and audio, with bounded temporary processing space). The UI is static on GitHub Pages; the API is a Docker image on an OVH VPS behind Caddy, bound to 127.0.0.1:8080.

Core entities: User, Profile, Friendship, Post, Comment, Event (plus occurrences and applications), Media, Conversation/Message, AuditEvent. Identifiers are UUIDs. Timestamps are UTC. Soft-delete keeps nested threads and moderation consistent. The wiki ER diagram explains the logical model; service Flyway migrations define the physical schema.

5. Algorithms

5.1 Friend suggestions

Two-stage generate-and-score. Candidates (at most 200): friends of friends, same city and sport, co-participants in the last 90 days; minus self, friends, pending, blocked, dismissed, locked. Score $S(u,v) = 0.35 m + 0.25 J + 0.15 G + 0.15 T + 0.10 E$, with m = Adamic-Adar on mutual friends, J = Jaccard of sports, G = geo ($D = 25$ km), T = window overlap, E = experience closeness. The card primary reason is argmax of weight times feature. Scores are stored in PostgreSQL and refreshed nightly or on demand. The scorer sorts at most 200 candidates in $O(C \log C)$; graph queries add cost that must be measured. Collaborative filtering was rejected: no implicit-feedback volume, and it cannot explain why.

5.2 Filtered search

People and events are two indexes with cursor pagination. Filters include sports, city, radius, remaining capacity, and friend-state. Private strangers never appear. The service loads catalog candidates from PostgreSQL, filters and ranks them in Java, then returns a stable page. This requires an $O(N)$ scan and $O(M \log M)$ sort. PostgreSQL full-text indexes are a future improvement, not the current implementation.

5.3 User matching

Weekly opt-in greedy assignment with a unique pair, no block edges, and a draft instant event of capacity 1 (private, with the peer invited) at the start of the overlapping window. Greedy is a 1/2-approximation; the wiki records that we do not show an empirical gap versus exact. Event accept order (FS-EVT-13) reuses the same matching score as a suggested queue, not as a capacity override.

6. Security

Access JWT is HS256 with a 15-minute lifetime. Refresh credentials last 14 days in an HttpOnly, Secure, SameSite=None, Partitioned cookie at /api/v1/auth. Redis atomically consumes rotated credentials. Passwords use Argon2id. Services enforce ownership, membership, visibility and staff permissions. Media reads receive 60-second signed URLs only after authorization. SeaweedFS is deployed behind an HTTPS object gateway. Real browser checks passed image upload, avatar update, event cover and audio playback on both accounts. The first registration on an empty database becomes admin; an operator CLI can provision staff on an existing database. Initial access therefore needs operator control.

7. Implementation notes

OpenAPI 3.1 in gym-buddy-openapi is the HTTP source of truth. The service generates Java models and API interfaces at build (controllers implement those interfaces; generated sources are not committed). The UI generates a TypeScript client with orval from node_modules/gym-buddy-openapi/openapi/openapi.yaml. Spring springdoc /v3/api-docs is never the published contract.

Focused branches and reviewed pull requests use one-line Conventional Commits. A verified issue may appear in the scope, for example feat(#59): add profile editing. Release tags use vMAJOR.MINOR.PATCH. Deployment must build the exact tagged source and record the image revision. The browser uses the HTTPS API at <https://vps-c39cdf03.vps.ovh.net/api/v1>. A tag or passing build alone does not establish that a repair is deployed.

8. Tests and fixtures

Backend unit tests name FS IDs (fsEvt07_concurrentLastSeatAcceptsExactlyOne, fsProf04_strangerOnPrivateProfileSeesStub, ...). Integration tests use Testcontainers. CI smoke hits GET /api/v1/healthz. Fixtures: Datafaker, seed 20260813, default 3 000 users / 12 000 requested friendships / 15 000 posts. CI verifies both small and 1 000-user datasets. The final backend CI passed 294 tests with no skipped integrations. Spring profile prod cannot reset. Ten stock object-storage objects are reused so the 5 000 media metadata rows do not fill the disk.

9. Critical analysis

Strengths: a demoable goal; a modular monolith that still shows Software Engineering structure; two distinct algorithms (explainable scoring vs constrained assignment); object storage that answers the brief's disk and ACL items; spec IDs that tests cite.

The September release was checked in separate member and staff browser sessions. Image/audio delivery, moderation, event acceptance and occurrence isolation passed. The database migration preserved 23 tables with content, schema and sequence comparisons. Scanned API, PostgreSQL, Redis and SeaweedFS images had zero CRITICAL findings; Caddy retains lower-severity advisories. Search scans candidates in Java; recommendation latency and relevance need fixture-scale measurement. Weekly matching uses a greedy approximation, and recurrence supports a weekly subset. Audio messaging and recurring events are required by the brief and cannot be dropped while claiming full compliance. See the dated release verification page for remaining gaps.

10. Conclusion

Software Engineering is evidenced by UML, Flyway, Gitflow, tickets on one board, and reviews against FS IDs. Web Technologies is evidenced by a versioned OpenAPI contract, JWT, WebSocket, Angular, and signed media. Algorithms is evidenced by FoF generate-and-score, filtered search, and weekly matching — implemented in Java, unit-tested, without a hidden SaaS.

11. Appendix

Released website evidence (version 1.2.0, 14 September 2026)

Figures below show the published member and staff applications using synthetic test accounts. Passwords and production configuration are not included. Historical August captures remain separately dated in the screenshot archive.

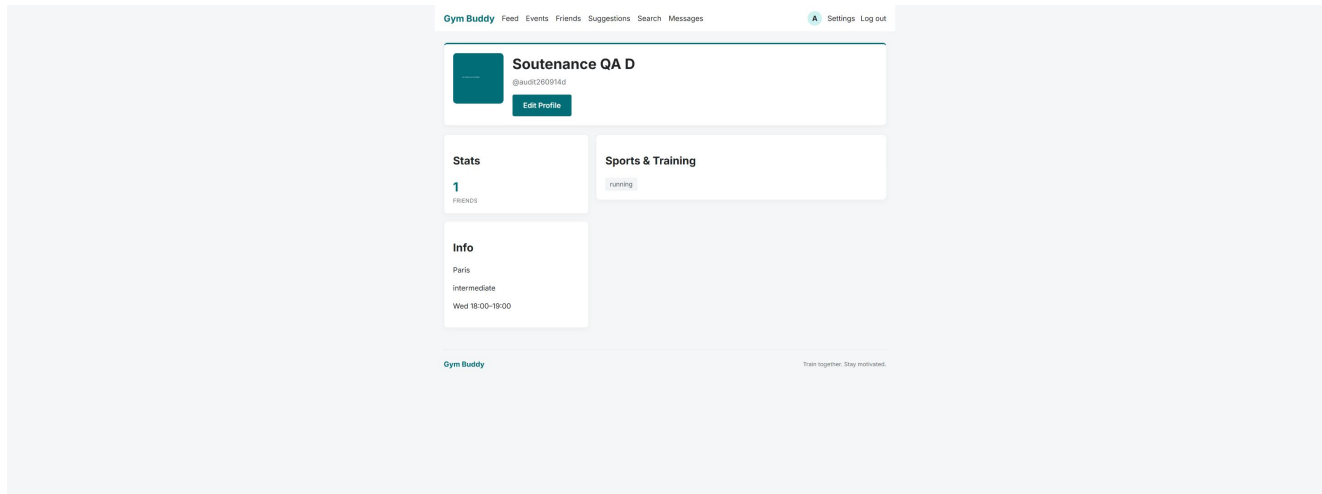


Figure 1. Saved profile avatar, rendered on the released member application.

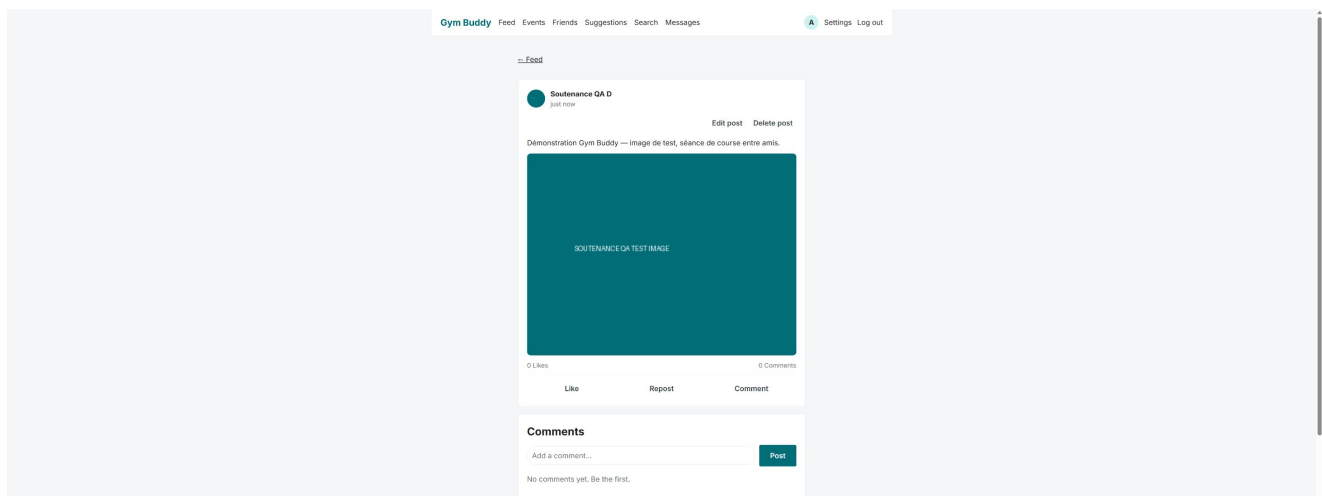


Figure 2. A synthetic member created an image post; its detail renders the processed image.

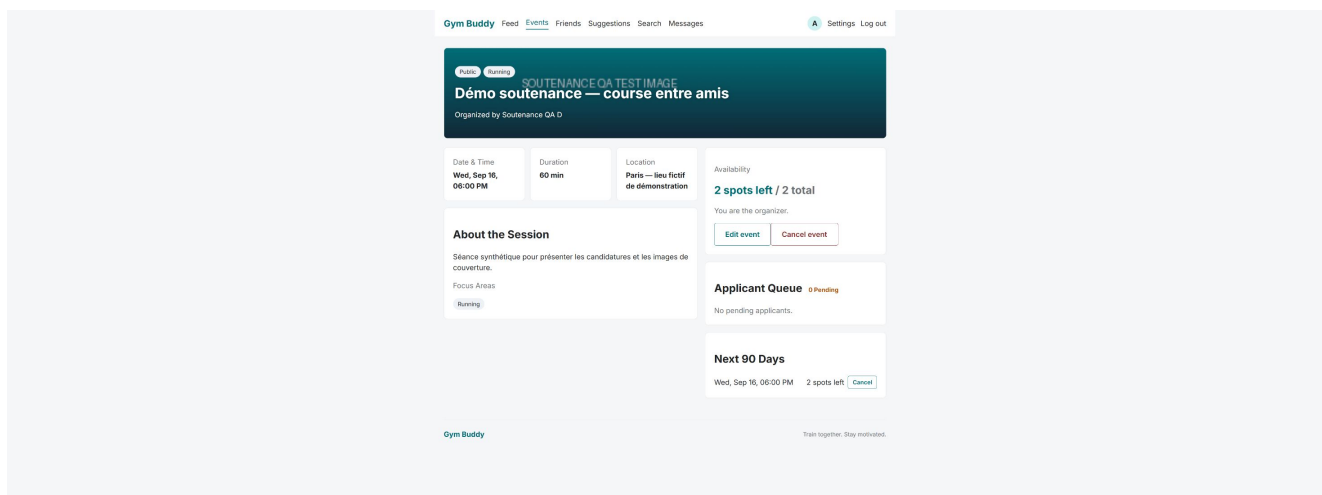


Figure 3. Public event with uploaded cover, title, location and capacity.

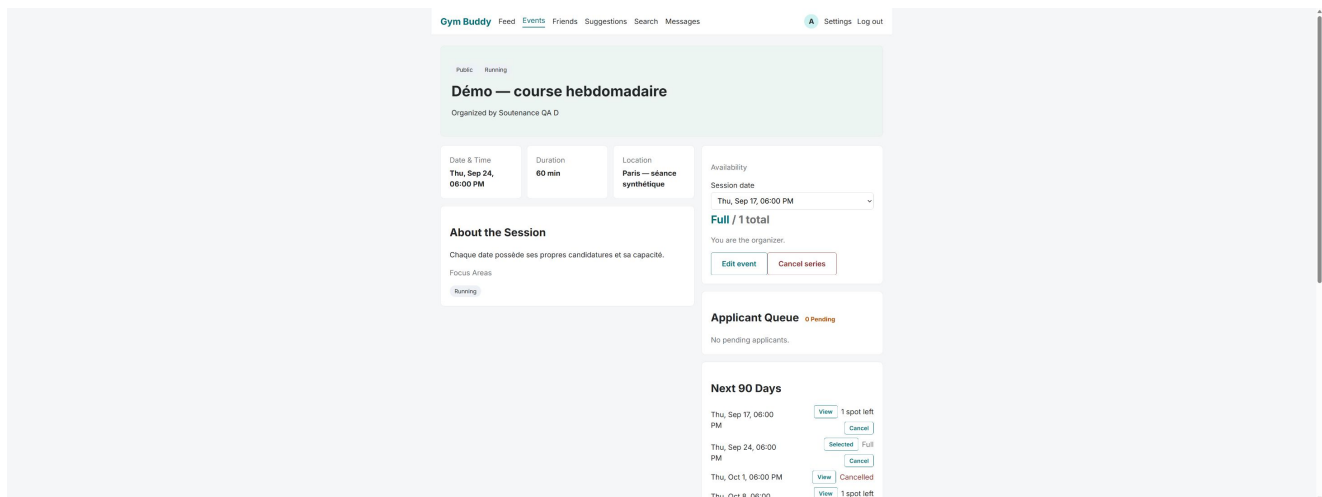


Figure 4. Applications and cancellation are isolated by weekly occurrence.

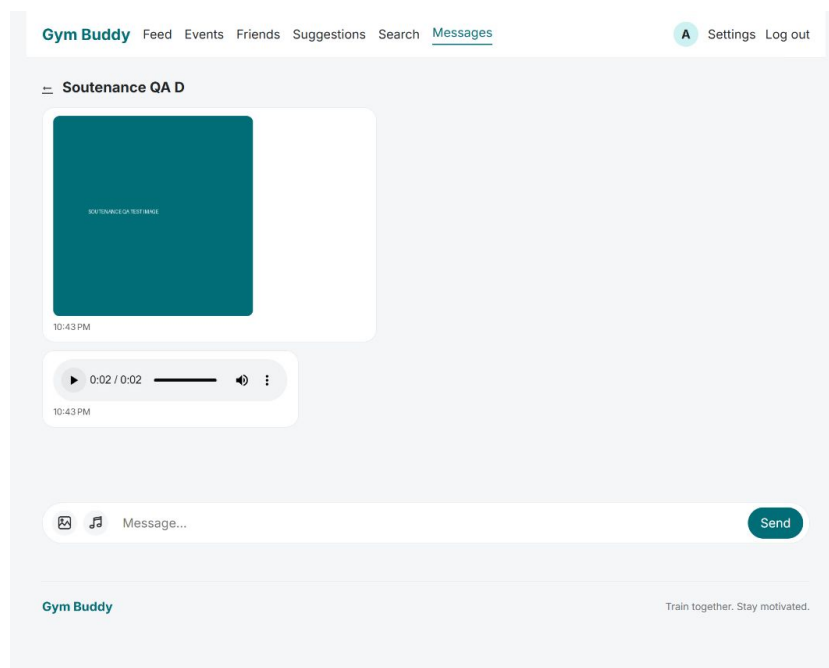


Figure 5. Image delivery without reload and audio playback between two synthetic friends.

Admin Portal

BACK OFFICE

Users

Content

Reports

Media

Fixtures

Audit

demo.admin

admin

Log out

User Management

Manage accounts, roles, and access across the platform.

Search

audit260914d

×

Role

All roles

Status

All statuses

Lock reason

Required to lock an account

User	Status	Joined	Role	Actions
Soutenance QA D audit260914d · audit260914d@example.invalid	active	2026-09-14	Member	Lock
Soutenance QA C audit260914c · audit260914c@example.invalid	active	2026-09-14	Member	Lock
Soutenance QA B audit260914b · audit260914b@example.invalid	active	2026-09-14	Member	Lock
Soutenance Audit Alex audit260914a · audit260914a@fixtures.gym.test	active	2026-09-14	Member	Lock

Entries: 4

Figure 6. Released staff user list filtered to the synthetic member accounts.

Admin Portal

BACK OFFICE

Users

Content

Reports

Media

Fixtures

Audit

demo.admin

admin

Log out

Audit Log

Append-only record of staff actions. Admin only.

Search log

demo.admin

×

Action

All actions

Timestamp	Staff	Action	Target	Reason
Sep 14, 2026, 10:37:45 PM	demo.admin	resolve report	report ba75050a-fc99-493c-9155-ff53693758cd	—
Sep 14, 2026, 10:37:38 PM	demo.admin	unhide content	post 94f1186c-87f4-47b2-8ba7-8190d9bd3b51	—
Sep 14, 2026, 10:37:34 PM	demo.admin	hide content	post 94f1186c-87f4-47b2-8ba7-8190d9bd3b51	Synthetic release verification: temporary hide and restore.
Sep 14, 2026, 9:30:05 PM	demo.admin	resolve report	report 350f1ed2-363b-4702-bb63-daad154b7216	—

Entries: 4

Figure 7. Audit records for synthetic moderation actions.

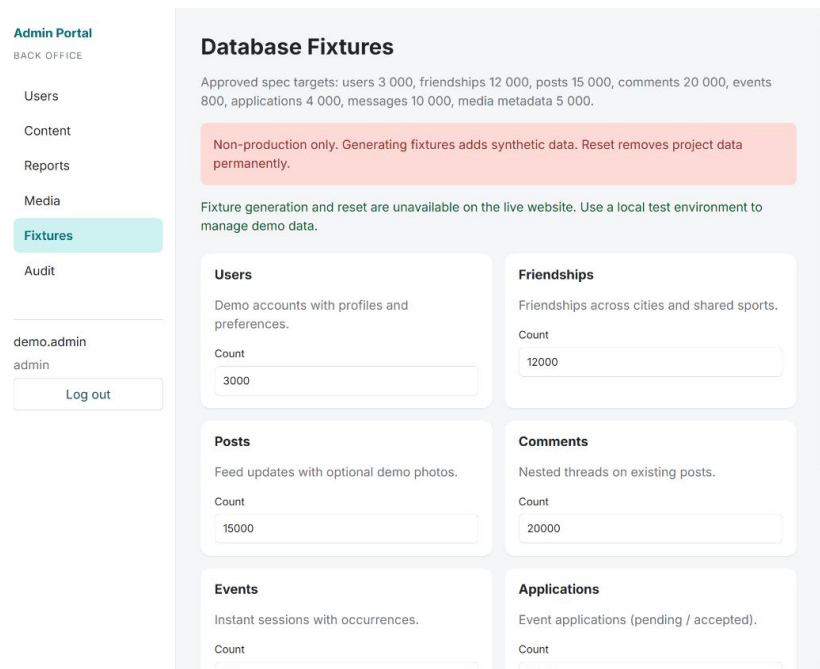


Figure 8. Fixture operations are explicitly disabled in production.

#	Shot	FS
1–2	Avatar and image publication	FS-PROF / POST
3–4	Event cover and occurrence isolation	FS-EVT
5	Image and audio conversation	FS-MSG / MED
6–8	Staff users, audit and fixture guard	FS-ADM

UML (use case, activity, sequence, class) is in 60-UML-diagrams. Selected HTTP surface is 40-Technical-specifications/09-Target-HTTP-surface.md.

The brief specified delivery to maurras.togbe@isep.fr by 31 August 2026. Working language of the wiki is English; the defense deck and rehearsal guide are in French. Source review: 1.2.0, 2026-09-14.